

Task2_3_Block1

Task2

1. Number 2.) In theory: Independent of its length. It is only determined by the geometry(a, b) and material parameters (L, C)

$$Z_0 = \sqrt{\frac{L}{C}}$$

In practice: Long lines introduce losses and parasitics, which can affect the measured impedance.

2. Number 3.) Yes. Wavelength λ strongly frequency dependent

$$\lambda = c/f$$

3. Number 5:)

$$Z_0 = \frac{60}{\sqrt{\epsilon_r}} \ln \left(\frac{b}{a} \right)$$

- $D_o = 6.66 \text{ mm}$

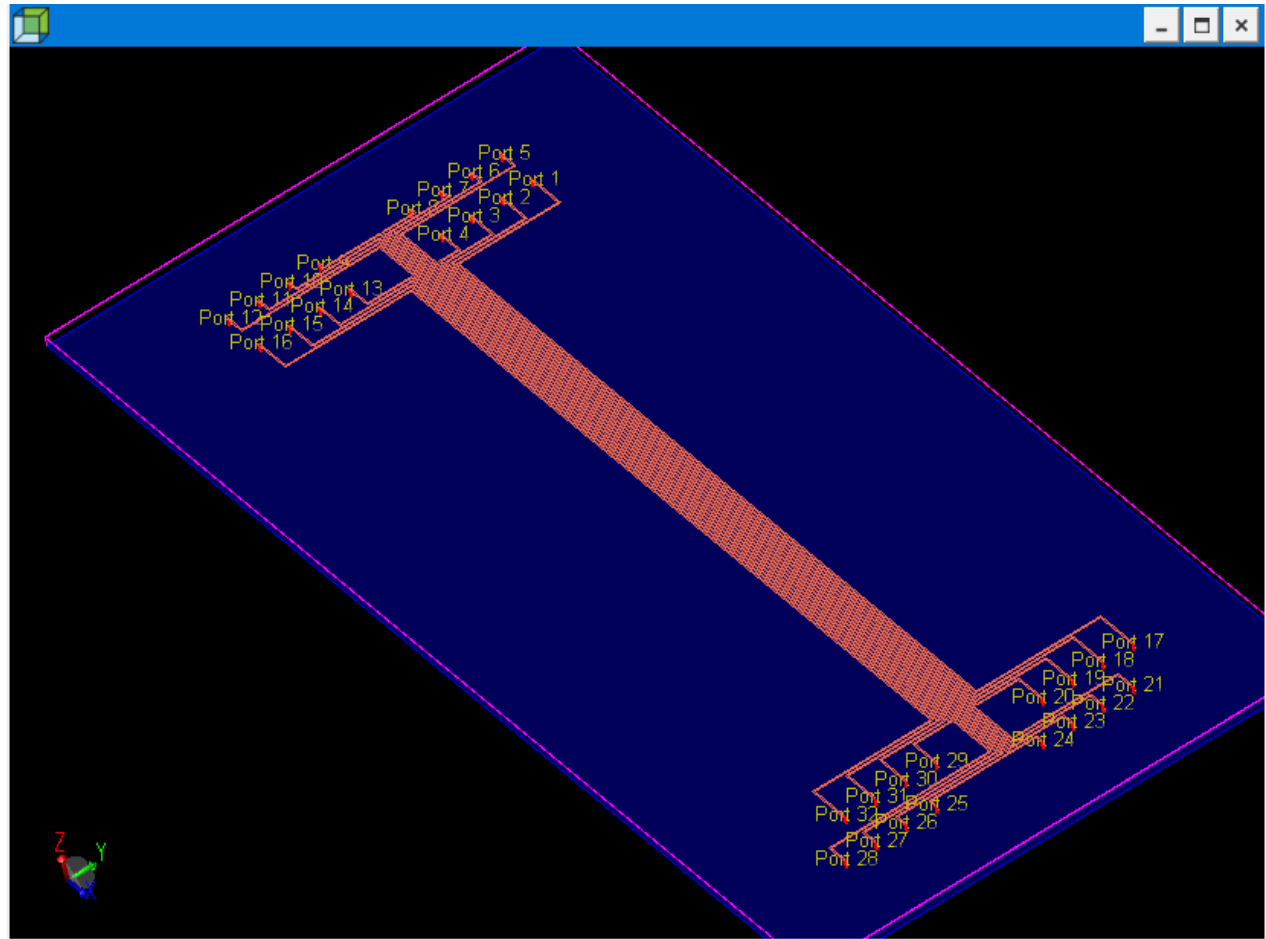
4. Number 6 :) Ground to Ground Spacing is $2W + G = 16 \mu m$

5. Number 8:) 70Ω : $W = 29.6 \mu m$

$$30 \Omega$$
: $W = 176.7 \mu m$

Task3

1. Number2:)



S-parameter Ports

Number	Gnd Layer	Name	Feed Type	Ref Impedance [Ohm]	Ref Offset [um]	Term Type
1	<Implicit>	P1	Direct	50 + 0i	N/A	inputOutput
		P1				
		Gnd				
2	<Implicit>	P2	Direct	50 + 0i	N/A	inputOutput
		P2				
		Gnd				
3	<Implicit>	P3	Direct	50 + 0i	N/A	inputOutput
		P3				
		Gnd				
4	<Implicit>	P4	Direct	50 + 0i	N/A	inputOutput
		P4				

Layout Pins

Name	Layer	Net	Connected to	Purpose	X [um]	Y [um]	Number	L
P1	cond	N_15	1(+)	drawing	-65989.2	-8255	1	1
P2	cond	N_14	2(+)	drawing	-65989.2	-13335	2	1
P3	cond	N_13	3(+)	drawing	-65989.2	-18415	3	1
P4	cond	N_12	4(+)	drawing	-65989.2	-23495	4	1
P5	cond	N_11	5(+)	drawing	-73609.2	-8255	5	1
P6	cond	N_10	6(+)	drawing	-73609.2	-13335	6	1
P7	cond	N_9	7(+)	drawing	-73609.2	-18415	7	1
P8	cond	N_8	8(+)	drawing	-73609.2	-23495	8	1
P9	cond	N_7	9(+)	drawing	-73660	-38735	9	1
P10	cond	N_6	10(+)	drawing	-73660	-43815	10	1
P11	cond	N_5	11(+)	drawing	-73660	-48895	11	1
P12	cond	N_4	12(+)	drawing	-73660	-53975	12	1
P13	cond	N_3	13(+)	drawing	-66040	-38735	13	1
P14	cond	N_2	14(+)	drawing	-66040	-43815	14	1
P15	cond	N_1	15(+)	drawing	-66040	-48895	15	1
P16	cond	N_0	16(+)	drawing	-66040	-53975	16	1

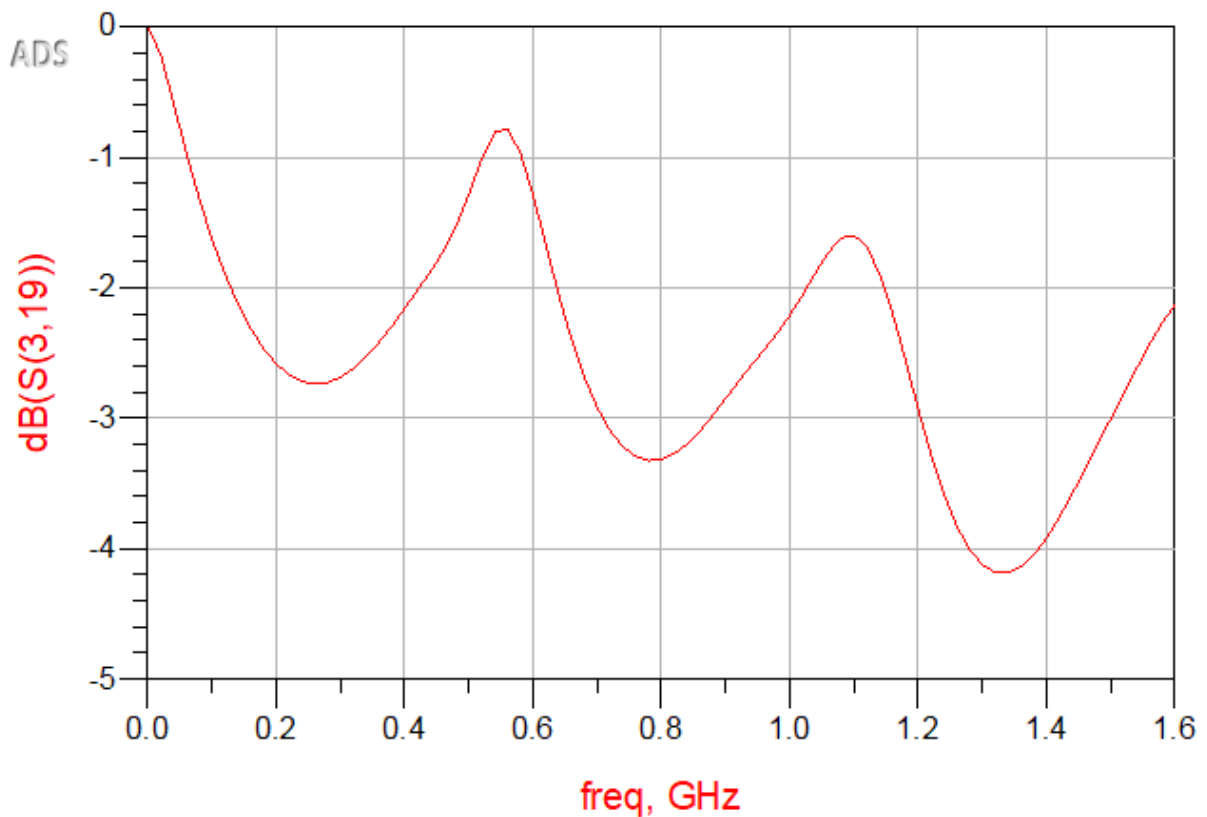
2. Number3:)

- Linecalc only calculates transmission line parameters
- Momentum performs full EM field simulation
- Structures: bends, junctions or coupled lines

3. Number4:)

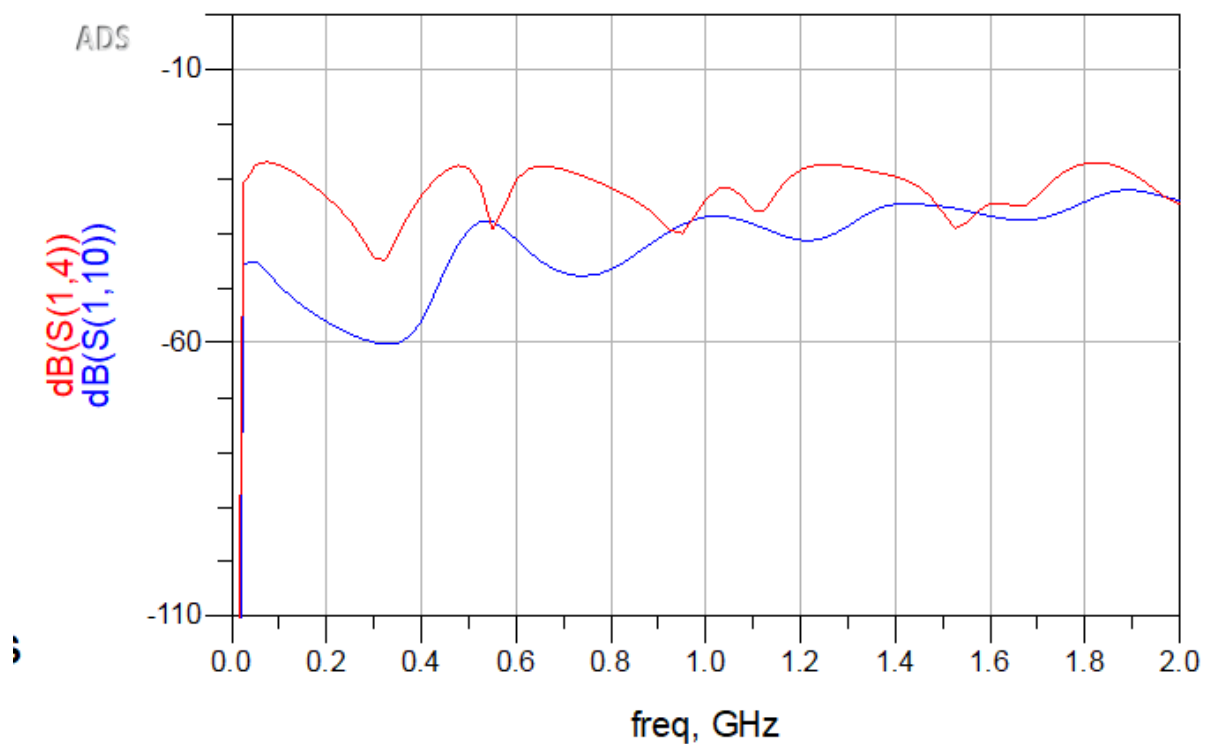
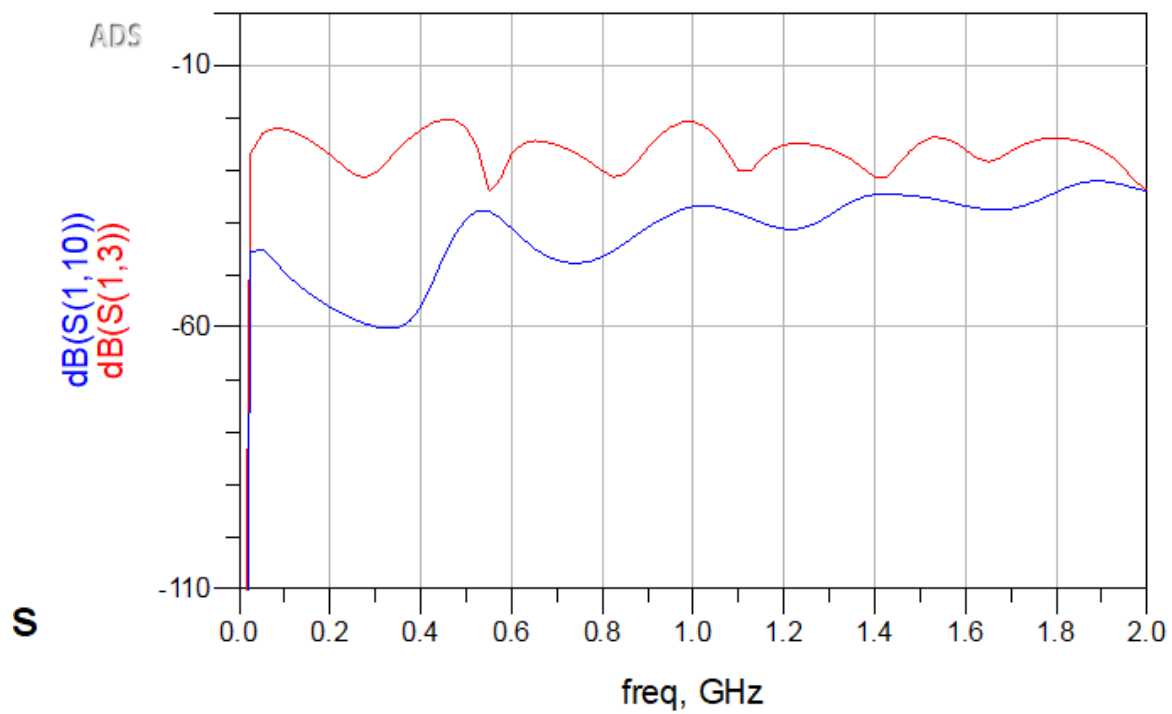
- Impedance transform
- Phase shift

4. Number5:)



- The resonating behavior is caused by standing waves due to reflections along the transmission lines
- Line length is a multiple of $\lambda/2$ (maxima)
- $\lambda/4$ (minima)

5. Number7:)



- $S(1,3)$ shows stronger crosstalk because the lines are physically closer and have stronger electromagnetic coupling
- $S(1,10)$ is weaker due to larger distance

6. Number8:)

$$P_{out} = P_{in} \cdot |S(10,1)|^2$$

$$S(10,1)_{max} = -32.15db$$

$$P_{out} = 6.1mW$$

7. Number9:)

- directional coupler
- high-speed circuit